CCCA Meta-Atlas — Cell-State Manifold Analysis

Liu et al., 2021 · Multiple Myeloma (Hematologic) · 10x · 24 samples / 74,986 cells

GO / NO-GO GATES

Gate 1 — integration & continuity — GO

Clonal MM plasma cells form a per-patient archipelago without integration (pre-Harmony iLISI 1.06). Judged on the regenerated embedding across a theta sweep (not a scalar): at theta=15 the 15 samples share ONE continuous manifold (island strip 42.5->28->11.7->2.4->2.7% as integration rises to a clean-speck level), well mixed per-neighbourhood (iLISI 2.27 of 5.62 ceiling; patient |r|<=0.19 on diffusion axes). Detached specks are unintegrated clones of the same states, not a distinct lineage.

Gate 2 — embryonic module pattern — GO (non-branching)

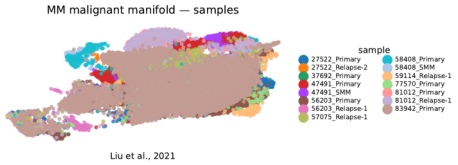
Single developmental axis: Gast & pSom vary along it with organised cross-repression (smoothed r = -0.67), and Crest occupies the opposite terminal end (pSom-Crest r = -0.95). No Gast/pSom co-expression hub and no opposing Gast arm -> non-branching (§1 organised cross-repression case). FLAG: Crest enriches the Gast end (Crest-Gast r = +0.56, Crest-pSom r = -0.95) — inversion of the canonical pSom-arm Crest tip; per §2 this is flagged but PASSES.

Gate 3 — Gast<->pSom anti-correlation — NA

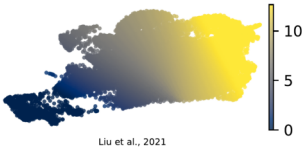
Non-branching manifold has no bifurcation; the tree-cluster cross-repression test does not apply.

STEP FIGURES

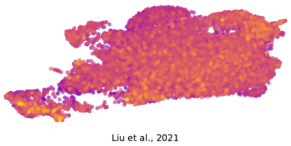
01 · Integration (samples)



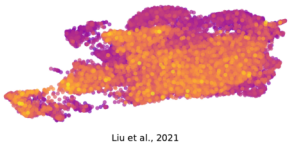
02 · Trajectory / pseudotime



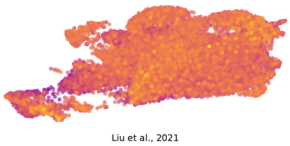
03 · Gastrulation



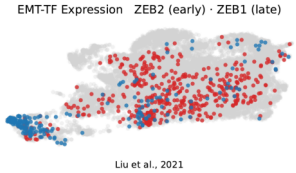
03 · Somitogenesis



03 · Neural Crest



04 · EMT-TFs (ZEB2>ZEB1)



Dataset. 17,267 malignant plasma cells; 15 marrows pass the 66->156 count cliff (>=100 cells; 17,003), 16,551 retained after removing detached islands. Global module means Gast -0.024, pSom +0.032, Crest -0.006 -> Linear-pSom/Crest class (pSom the strongest module).

Verdict. The embryonic-development pattern REPRODUCES as an organised non-branching cross-repression axis (Gast/pSom early, Crest terminal), with a Crest-on-Gast-arm inversion flag. Gates: GO / GO / NA.

PARAMETER CHOICES & RATIONALE

Min-cell threshold >=100 cells/sample (drops 5 thin marrows)
Lower-half count cliff at 66->156 (ratio 2.4); thinner marrows cannot seed a stable 30-NN nbhd.

Integration (Harmony) theta = 15 on sample, 20 PCs
Strongest integration where iLISI plateaus AND the island fraction bottoms out; clonal MM needs it.

Embedding PCA-50, kNN=30, UMAP spread 1.3
10x-scale precedent (Kuerten/Galen), re-derived not inherited from Puram (40 PCs).

Geometry Non-branching principal curve
PPT-tree falsification shows no Gast/pSom hub / opposing arm; ElPiGraph curve fold-free (SA 2.9).

Scope Whole manifold (no lobe excised)
Developmental poles span both cycling & low-cycling compartments; axis orthogonal to proliferation.

Root Gast/pSom-high, Crest-low tip
Developmental order: gastrulation/somitogenesis EARLY, neural crest LATE.

EMT-TF panel SNAI/ZEB/TWIST + FOXC/PRRX, >3% floor
Only ZEB2 (early) & ZEB1 (late) clear the detectable-expression gate; VIM excluded (not a TF).

Determinism: Harmony KMeans is non-deterministic; iLISI +/-0.2 and pseudotime-ordering rho vary run-to-run. The manifold geography (smoothed module correlations) and all three gate calls are stable across reruns. Seed = 1; scFates 1.1.0 / scanpy 1.11.1.